IC next-10 item 05: multi-polynomial Coppersmith reach

Item 05: Coppersmith reach with several polynomials and shared targets

Date. 2026-10-14. **Campaign.** `research/ic_next10_20261014/` item 5 ("### 5." in `reports/Index calculus leads next steps.md`). **Protocol.** `PROTOCOL.md`, frozen before any data. **Code.** `examples/ic_next10_05_multipoly_lattice.rs` (Rust: the exact min-cut optimiser and the LLL runs), `milp_multi.py` (MILP over all monomial sets and all leading vertices), `asymptotic_box.py` ($\ell \to \infty$ box family), `plot_05.py`. **Run.** `reports/run-20261014T1905Z/`.

Verdict: NEGATIVE

None of the variants reaches a δ at which relation collection beats rho.

For S_5 (arity m = 4), the best variant is three targets sharing two summands. Its asymptotic reach is **$\delta^* \approx$ 0.01398**, which is 1.12 × the single-polynomial 1/80 = 0.0125. The best two-target variant reaches 0.01361 (s = 3) and 0.01313 (s = 1).

Every variant whose leading monomials are top corners stays **below the unchanged ceiling 1/32 = 0.03125**. That ceiling, proved in §3, holds for every monomial set, every multiplicity ℓ and every sublattice. The report's kill criterion is "an LP δ^* for S_5 below 1/32 in every new family". It is met.

- **Shared targets.** The economics would rule them out even with a perfect oracle. With n small unknowns and r equations, relation collection beats rho only when $(r - \frac{1}{2})/(n - 1) < \delta < 1/4$. At m = 4, every sharing pattern (k ≥ 2 targets, s ≥ 1 shared summands) gives $(r - \frac{1}{2})/(n - 1) \geq 1/4$, so this window is empty. The best such pattern only ties rho, with exponent 1/2.
- **Curve equations.** Adding them gives **exactly no gain**. The exact MILP picks the bare S_3 lattice for every leading-vertex choice: 9/100 at $\ell = 1$ (8 + 16 choices) and 13/112 at $\ell = 2$ (8 choices).
- **Semaev chain with large intermediate unknowns.** It gives **no lattice at all** ($\delta^* = 0$).

The one variant with a real reach gain is the shared-target S_3 pair. At toy size LLL confirms the gain is small. At 56 bits and $\delta = 0.13$, the pair recovered the root in 7 of 12 trials against 2 of 12 for the single S_3. At 40 bits the counts were 17 of 24 against 11 of 24. Fisher p-values are 0.09 and 0.14.

The positive control reproduces W10 exactly:

S_3, ℓ = 1 … 12	**S_4, ℓ = 1, 2**	**box limits**
9/100, 13/112, 29/225, 3/22, 90/637, 335/2312, 959/6498, 22/147, 1761/11638, 229/1500, 407/2646, 202/1305	1331/70875, 79/3087	1/6, 1/24, 1/80 (numeric to 1e-12)

The null control also behaves as expected: two S_3 on disjoint unknowns give exactly the single-S_3 rational at every $\ell = 1 \ldots 6$.

1. Question and prior work

W10 (`research/lit2026_w10_s5_coppersmith_20261011/`) solved the exact LP for full-rank triangular single-polynomial lattices. It found δ* = 1/6, 1/24, 1/80 for S_3, S_4, S_5, with a family ceiling of `1/(md)` = 1/4, 1/12, 1/32. W10 §7 left four routes open:

- several polynomials (curve equations, shared unknowns);
- sublattice strategies;
- unravelled linearisation;
- Gröbner-optimal shifts.

Prior-work check: grep over `research/` and `benchmarks/` found nothing beyond W10 (PROTOCOL.md). `conductor check` failed with connection refused, and the work continued per RULES.md.

2. Family and exact optimisers

The unknowns are `x_v` with `|x_v| < p^{λ_v}`, where `λ_v = δ` for a small unknown and `λ_v = 1` for a large one (a curve `y`, an intermediate Semaev point `t`). The polynomials `f_1..f_r` are taken mod p. Shifts have the form `p^{ℓ−|k|} x^γ ∏ f_j^{k_j}`, with leading monomial `β = γ + Σ k_j a_j`. Here `a_j` is a vertex of `Newt(f_j)`, and all `a_j` are simultaneous unique maxima of one weight, so every product is triangular. There is one shift per leading monomial, and its support lies in `M`. As p → ∞, success means `Σ_{β∈M} |k(β)| > Σ_{β∈M} λ·β`. This is the direct generalisation of W10's family `F`, and it reduces to `F` when r = 1.

1. **Rust, exact, down-closed `M` with top-corner leading monomials.** With box supports and top corners, every shift at β has support inside the box `[β − dK, β]`. So for down-closed `M` the best gain is `k(β) = max{|k| ≤ ℓ : Σ_{j∋v} k_j d_{jv} ≤ β_v}`. The optimum over down-sets is a max-weight closure, solved by Dinic on symmetry orbits inside Dinkelbach. Each δ_ℓ is the exact rational ratio of the returned `M`. No `M` beats it by `2^-30` or more, because the next Dinkelbach solve, just above it, returns an empty closure. The bounding box grows until it no longer binds.
2. **MILP over all `M` and all leading vertices** (`milp_multi.py`, HiGHS). Binary `m_β` and `z_{β,k}` with `z ≤ m_{β'}` for `β'` in the support, and `Σ_k z_{β,k} ≤ 1`. Every vertex tuple admitting a common weight is tried. This covers non-box supports (the curve equation $y^2 - x^3 - ax - b$), non-top leading monomials, and non-down-closed `M`.
3. ℓ → ∞ **box family** (`asymptotic_box.py`). In scaled coordinates the gain fraction is `g(u) = min(1, min_S u, Σ_i min_{A_i} u)` and `δ = E[g]/(d E[Σu])` over per-group boxes. E[g] is computed by 1-D quadrature with an FFT-convolved sum-of-minima law, and the result is checked against a closed form to 6e-7. For the single polynomial this gives `2/(m(m+1)d)` to 1e-12. The finite-ℓ optima are **non-box** down-sets for the pair, so the box value is a lower bound on the down-set δ*. For the S_3 pair, the finite-ℓ sequence extrapolates to 0.1808, while the box value is 0.1779.

3. Ceiling for every M, every ℓ and every sublattice (top-corner leading monomials)

The leading monomial of `x^γ f^k` is `β = γ + Σ k_j a_j ≥ Σ k_j a_j` componentwise. Hence `|k(β)| ≤ G(β) := max{|k| : Σ k_j a_j ≤ β}`. If `λ·a_j ≥ 1` for every j, then for every β, `λ·β ≥ Σ k_j λ·a_j ≥ |k|`, so `Σ_M |k| ≤ Σ_M λ·β` and the lattice fails.

Sublattices. The basis is triangular in the monomial coordinates, with scaled diagonal `D_β = p^{ℓ−k(β)} X^β`. Its k-th exterior power is triangular with diagonal products, and λ_1 of a lattice is at least its smallest diagonal entry. So any rank-D′ sublattice has determinant at least the product of the D′ smallest `D_β`. The Howgrave-Graham condition `det^{1/D′} < p^ℓ` therefore again needs some β with `k(β) > λ·β`. This covers Boneh–Durfee-style sublattices.

Ceiling. Success needs **some** polynomial in the system with $\lambda \cdot a_j < 1$. With only small unknowns the ceiling is $\max_j 1/\deg_tot(f_j)$, which is $1/(md)$ for every Semaev-based system. Adding polynomials or sharing unknowns does not raise it. A polynomial that contains a large unknown in its top corner (curve equation with y^2 leading, chain link with t^2) can never be the one that satisfies the condition.

Known-answer control: for Boneh–Durfee, the ceiling is $\delta + \frac{1}{2} < 1$, i.e. $\delta < 1/2$, and the literature value 0.2929 lies under it.

Not covered by this argument.

- Non-top leading monomials. These are covered at small ℓ by the MILP (§4).
- Gröbner-optimal shifts that cancel leading terms. With shared variables, $LT(I^j)$ can exceed the products of leading monomials.
- Unravelled linearisation. A substitution $u = q(x)$ with $U \approx X^{\deg q}$ leaves the top-vertex weight at 32δ when q has small integer coefficients. This is an argument, not a theorem, and a q with large coefficients is not small. W10's e-encoding is such a linearisation and does worse.

4. Results

4.1 Exact finite-ℓ optimum (Rust; `lp.md`, `lp.json`)

system	n small, n large, r	$\ell = 1$	$\ell = 2$	$\ell = 3$	largest ℓ run	vs single at same ℓ
C1 S_3	2, 0, 1	9/100	13/112	29/225	202/1305 = 0.154789 (ℓ=12)	= W10
N1 two S_3, disjoint	4, 0, 2	9/100	13/112	29/225	335/2312 (ℓ=6)	**identical** (null)
V3 S_3 pair, s = 1	3, 0, 2	9/95 = 0.094737	13/105 = 0.123810	122/885 = 0.137853	94/561 = 0.167558 (ℓ=12)	1.053, 1.067, 1.070, 1.083
V3 S_3 triple, s = 1	4, 0, 3	27/280	249/1988	1776/12727	3324/20995 = 0.158323 (ℓ=6)	1.071 … 1.093
C1 S_4	3, 0, 1	1331/70875 = 0.018780	79/3087 = 0.025591			= W10
V3 S_4 pair, s = 1	5, 0, 2	2662/141645 = 0.018793	474/18515 = 0.025601 (box binds: lower bound)			1.0007, 1.0004
V3 S_4 pair, s = 2	4, 0, 2	675/33488 = 0.020156	326/11737 = 0.027775	39/1225 = 0.031837		1.073, 1.085
V2 S_4 chain $S_3(x_1,x_2,t)$, $S_3(x_3,t)$	3, 1, 2	**0**	**0**	**0**		no lattice

Least-squares extrapolation $\delta_\ell \approx \delta_\infty - c/\ell + e/\ell^2$ over $\ell = 8 \ldots 12$ (derived, not exact): single $S_3 \to 0.16668$ (true 1/6); S_3 pair $\to$ **0.1808**.

4.2 Asymptotic box family ($\ell \to \infty$; `asymptotic_box.md`)

The economics column: with n small unknowns and r equations, the cost is $p^{\max(r-(n-1)\delta, 2\delta)}$. This beats rho iff $(r-\frac{1}{2})/(n-1) < \delta < 1/4$.

system	n	r	δ*_box	× single	ceiling	IC needs δ >	window non-empty?
single S_3	2	1	1/6	1	1/4	1/2	no
S_3 ×2, s=1	3	2	0.177886	1.067	1/4	3/4	no
S_3 ×3, s=1	4	3	0.182015	1.092	1/4	5/6	no
single S_4	3	1	1/24	1	1/12	1/4	no (tie)
S_4 ×2, s=1 / s=2	5 / 4	2	0.044171 / 0.045217	1.060 / 1.085	1/12	3/8 / 1/2	no
S_4 ×3, s=1 / s=2	7 / 5	3	0.045102 / 0.046427	1.082 / 1.114	1/12	5/12 / 5/8	no
single S_5	4	1	**1/80 = 0.0125**	1	**1/32**	**1/6**	**yes** (best δ = 1/5, exponent 2/5)
S_5 ×2, s=1 / 2 / 3	7 / 6 / 5	2	0.013134 / 0.013583 / 0.013610	1.051 / 1.087 / 1.089	1/32	1/4 / 3/10 / 3/8	no (s=1 ties rho at δ = 1/4)
S_5 ×3, s=1 / 2 / 3	10 / 8 / 6	3	0.013370 / **0.013982** / 0.013971	1.070 / **1.119** / 1.118	1/32	5/18 / 5/14 / 1/2	no

4.3 All monomial sets and all leading vertices (MILP; `milp_*.md`, `milp_*.json`)

system	ℓ	leading-vertex tuples	best δ_ℓ	note
C1 S_3	1	4 / 4	9/100 for every vertex	= W10 (all classes equal)
V3 S_3 pair, all `M`	1	4 of 16 finished	9/95, the same as the down-closed optimum, for 3 tuples; 1/11 for the constant/constant tuple, where the MILP hit its time limit (lower bound)	the other 12 tuples were stopped for time (§6)
V1 S_3 + curve eq. for x_1	1	8 / 8	9/100	optimum never uses the curve equation
V1 S_3 + both curve eqs	1	16 / 16	9/100	same
V1 S_3 + curve eq. for x_1	2	8 / 8	13/112	= single S_3 at ℓ = 2 (W10); curve equation never used (`milp_v1_l2.md`)
V2 S_4 chain (t large)	1	16 / 16	0	no `M` has positive gain even at δ = 0

For V1 with one curve equation, the “binds” flag comes from the unused coordinate y_2, whose bounding box is `[0, 0]`. It is an artefact of the flag, not a binding box.

4.4 LLL at toy size (the only variant with a gain: the S_3 pair)

Setup: closure-optimal $\ell = 1$ lattices. The single S_3 lattice has dim 25 and LP $\delta_1 = 0.0900$; the pair lattice has dim 45 and LP $\delta_1 = 0.0947$. Primes: `p = 1099511627563` (40-bit) and `p = 72057594037927931` (56-bit), the largest p ≡ 3 mod 4 below 2^bits; a = −3, b = 31. Planted points have `x < ⌊p^δ⌋`. LLL is the repo’s `lll_reduce` (f64 Gram–Schmidt, used up to dim 30) or `lll_reduce_hp` (above dim 30), with δ_LLL = 0.99. A trial succeeds when at least n reduced rows vanish over Z at the root. Measured values:

δ	40-bit single	40-bit pair	Fisher p	56-bit single	56-bit pair	Fisher p
0.12	19/24	24/24	0.05	9/12	11/12	0.59
0.13	11/24	17/24	0.14	2/12	7/12	0.09
0.14	7/24	9/24	0.76	0/12	1/12	1.0
0.15	6/24	4/24	0.72	0/12	1/12	—
0.16	1/24	2/24	—	0/12	0/12	—

Both lattices succeed above their $p \to \infty$ δ_1, which is the finite-p effect W10 also saw. The pair moves the 50% point by about +0.01 in δ, consistent with the LP’s +5%. No configuration recovers roots at $\delta \geq 0.18$. A pair $\ell = 2$ lattice (dim 224) was not run: at the measured 2–5 s per dim-45 hp reduction it would take hours per trial on the shared host.

5. Interpretation

1. Sharing unknowns does raise the exponent, but only by 5–12%, never by the ≥ 1.5× that would trigger a measured probe. The ceiling does not move.
2. Curve equations and chains add unknowns of size p. Every monomial in such a variable costs a whole p. The exact optimiser never uses them (V1), and when they are forced in (V2) no lattice exists.
3. For m = 4, the needed $\delta \approx 1/5$ is 16× above the single-polynomial δ* and 14.3× above the best multi-polynomial δ* (0.01398). The proved ceiling 1/32 is still 6.4× short. Shared-target variants also face an empty economic window.

Kill criterion (protocol). S_5 $\delta^* < 1/32$ holds in every new family. **The lattice route to m = 4 is closed for these families.**

6. Deviations from the protocol

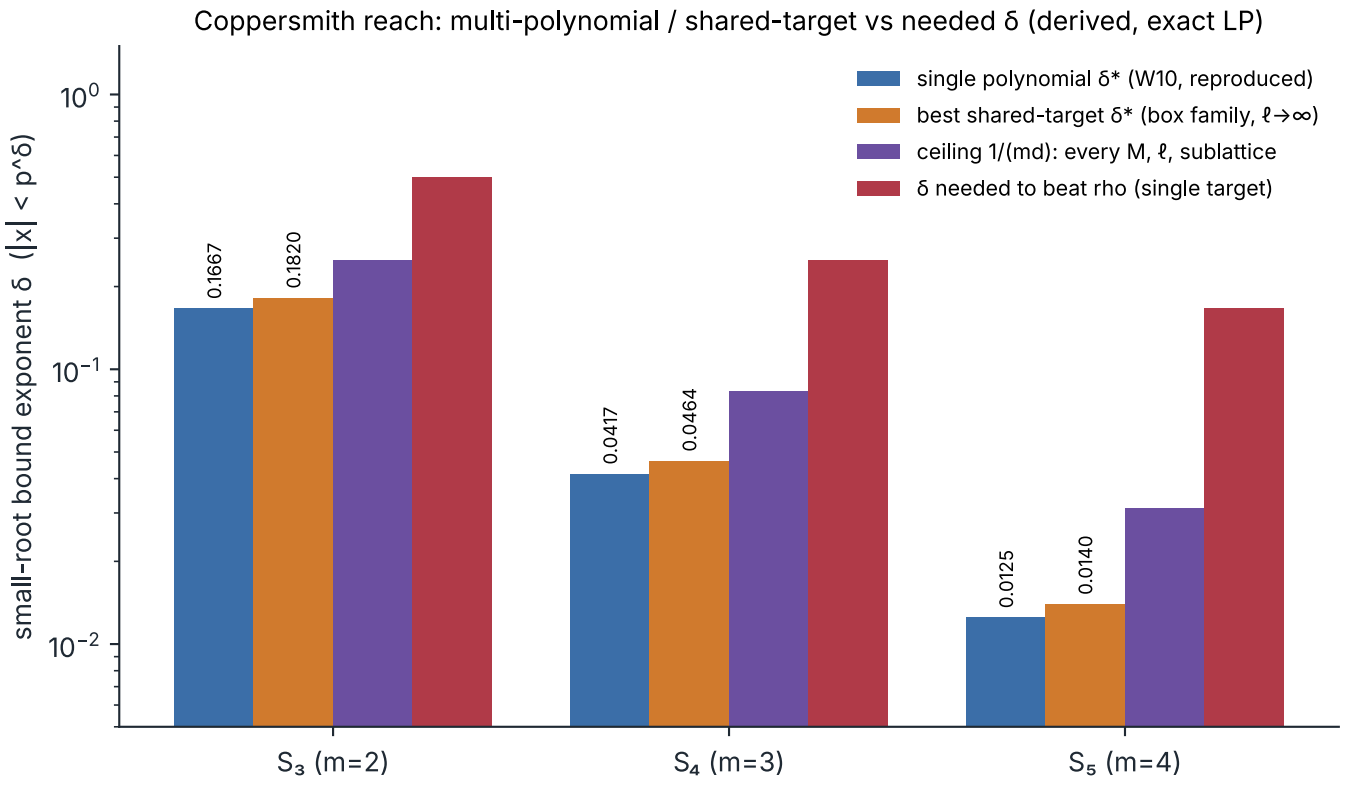
- The asymptotic and MILP numbers come from Python (scipy, HiGHS). RULES.md lists these tools, but AGENTS.md prefers Rust for results. Where they overlap, they are cross-checked against the exact Rust optimiser: C1 = 9/100 and pair = 9/95. The box quadrature also matches its closed form.
- The Rust binary that ran is the 18:12 build of `ic_next10_05_multipoly_lattice.rs` without the later `--only`/`--ells` flags. Those flags were added to the source, but the rebuild was blocked for over 30 minutes behind other items’ cargo builds. The output is identical, because the flags only filter cases.
- MILP over all `M` for the S_3 pair finished 4 of 16 leading-vertex tuples. The constant-vertex tuple hit its 400 s time limit. The sweep was stopped to free the CPU for V1/V2.
- MILP at $\ell = 2$ for S_3 alone (box 9×9) did not finish within 25 minutes and was dropped.
- LLL ran at $\ell = 1$ only ($\ell = 2$ pair dim 224 is infeasible with the BigInt hp LLL here). The 56-bit run used 12 trials per point, and the 40-bit fine grid used 24.
- S_5 systems were not solved at finite ℓ (7 unknowns, ≥ 10^9 orbit nodes). They are covered by the box asymptotics and the proved ceiling.

- The host was shared (load 25 on 4 cores). All timings are informational.

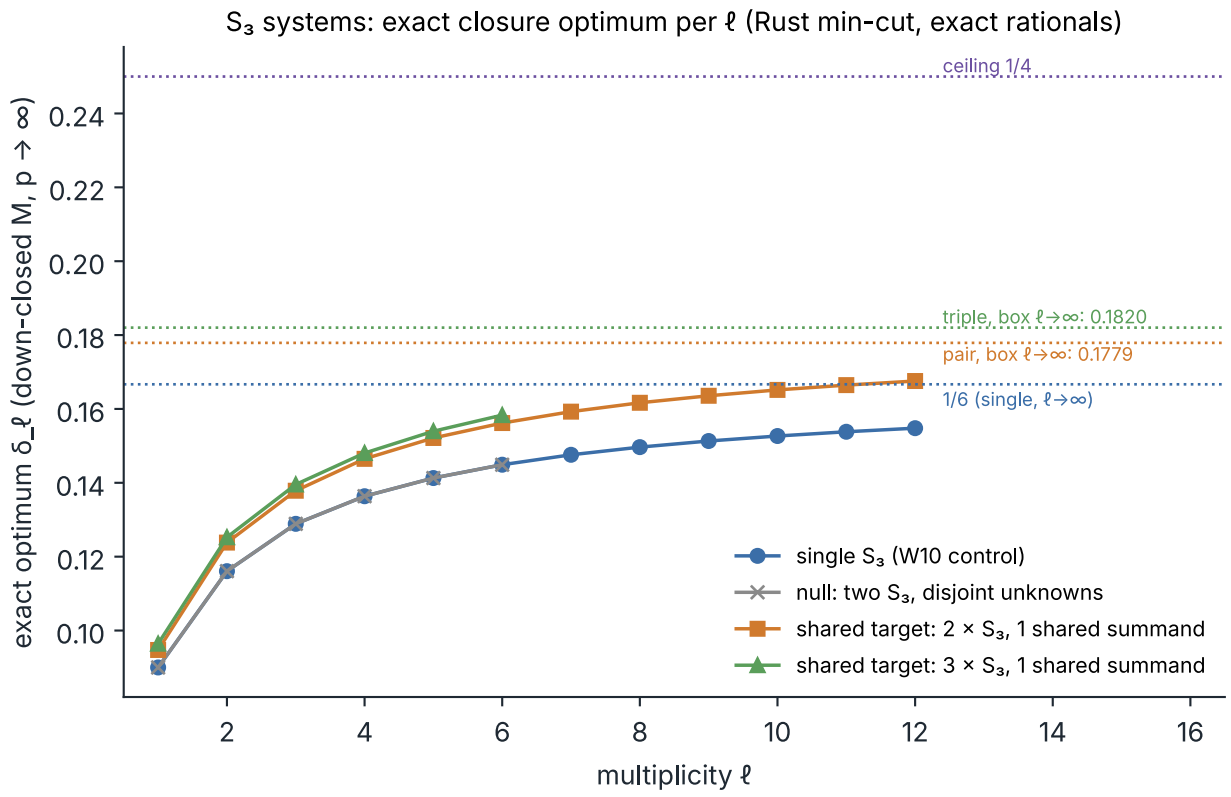
7. Reproduce

```
        cargo build --release --example ic_next10_05_multipoly_lattice
        B=target/release/examples/ic_next10_05_multipoly_lattice;
R=research/ic_next10_20261014/05-multipoly-lattice
        nice prlimit --as=734003200 -- $B --part lp --max-ell 12 --json $R/reports/run-
20261014T1905Z/lp.json
        nice prlimit --as=734003200 -- $B --part lll --bits 40 --ell 1 --trials 8  --json
.../lll40_l1.json
        nice prlimit --as=734003200 -- $B --part lll --bits 40 --ell 1 --trials 24 --deltas
0.11,0.12,0.13,0.14,0.15,0.16 --json .../lll40_l1_fine.json
        nice prlimit --as=734003200 -- $B --part lll --bits 56 --ell 1 --trials 12 --deltas
0.08,0.10,0.12,0.13,0.14,0.15,0.16,0.18 --json .../lll56_l1.json
        cd $R; python3 -I asymptotic_box.py asymptotic_box.json > asymptotic_box.md
        python3 -I -u milp_multi.py milp_c1.json c1; python3 -I -u milp_multi.py
milp_v1.json v1
        python3 -I -u milp_multi.py milp_v2.json v2; python3 -I -u milp_multi.py
milp_v3.json v3
        python3 -I -u milp_multi.py milp_v1_l2.json v1l2
        python3 -I plot_05.py reports/run-20261014T1905Z
```

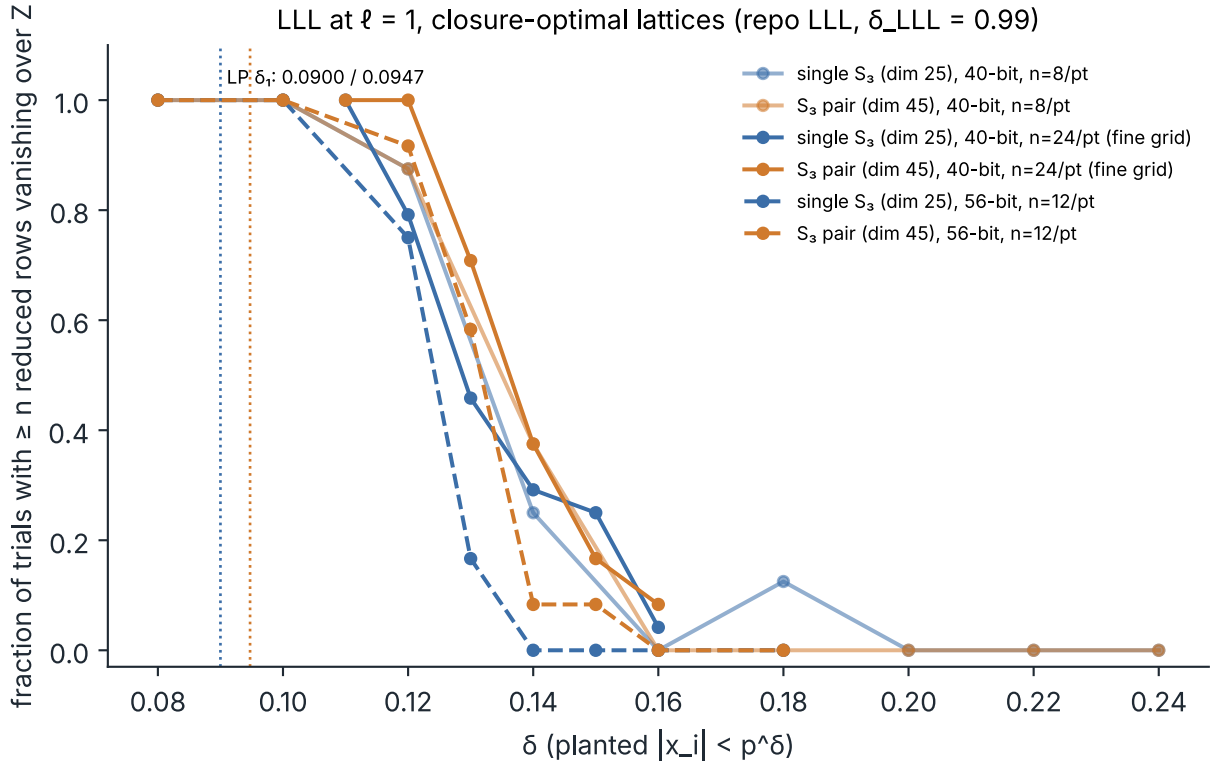
8. Graphs



- reach_vs_family.svg: single δ*, best shared-target δ*, ceiling and needed δ for m = 2, 3, 4 (log scale).



- finite_ell.svg: exact δ_ℓ against ℓ for the S₃ systems, with the null control overlapping the single polynomial.



- lll.svg: LLL success fraction against δ.

Canonical graphs checked and left unchanged: `docs/index-calculus-scoreboard.html`, `docs/ic/progress-timeline.json`, `docs/performance-gains/`. This is a feasibility negative with no speed result, so there is no `benchmarks/` copy.